

Developing Students' Entrepreneurial Skills Through E-TShirt Project

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Abstract

Economic conditions of many countries have not yet improved after the global recession, which hit the world in 2007. This has resulted in high unemployment rate in almost all countries of the world. One way to tackle this situation is to equip students with both business and technical expertise so that they can setup their own enterprise or start up a business in a crisis like situation. This paper demonstrates how university students were abreast with entrepreneurial skills through e-TShirt project in a course. Another aspect of this paper is to highlight the learning achievements of the students during this endeavor.

Developing Students' Entrepreneurial Skills Through E-TShirt Project

Introduction

Economic conditions of many countries have not yet improved after the global recession, which hit the world in 2007. It is a source of increase in poverty because of huge job losses. New graduates including the IT graduates are finding it difficult to get jobs. According to a survey (Hoffman, 2010), CS graduates are struggling for employment even in highly developed countries like UK and China. . Economic slowdown caused large number of layoffs in US IT sector at the height of the current economic turmoil. Thus, universities producing IT personnel must realize this bleak situation. Instead of just imparting technical knowledge to the students, they must also equip their students with business and entrepreneurial skills. This can help them to become self employed.

Albeit, some of the universities have started the entrepreneurial discipline but IT students cannot benefit from it. This is because their curriculum is already loaded with a long list of technical courses. Thus, it is difficult for them to create room for courses from other disciplines. In this scenario, the good option is to transform IT courses so that they have a flare of business and provide opportunities to students to develop entrepreneurial skills. Class projects can act as a vehicle to achieve this goal. This can also avoid drastic changes in the curriculum. Teachers can assign saleable/ marketable projects to the students. The entrepreneurial practices form the backbone of these projects. The research presented in this paper demonstrates how IT students can be accustomed to the business and entrepreneurial skills using class projects.

The research is conducted using the Logic Design and Switching Theory (LDST) course, which is taught to the 3rd semester students of Computer Engineering Department (CED) at (university name, country name). First author taught the course in the fall semester of 2010. Apart from disseminating business and entrepreneurial insight, this work also evaluates the students for their learning skills. E-TShirts act as the project ingredient in this research. Mentor divided the students into groups and asked them to develop an e-TShirt having a unique design. Objective was to use these innovative and attractive costumes as a source of business. A questionnaire, designed to collect data about business and *students'* learning related issues, provides the basis of data collection for this study. The remaining portion of this article describes the definition of important terms, previous work, E-TShirt project and exhibition, research methodology, results and the conclusion.

Business Activity, Entrepreneurship, and Entrepreneurial Insight

The aim of *business activity* is mainly to earn profit by satisfying customer needs. Most important aspect of business activity is that it helps in the growth and development of businesses, which in turn help in economic development as indicated by Hofstrand (n.d.). Thus, economic development, which aims at creating jobs, is actually a by-product of business activity.

The person who manages and organizes the business for the sake of earning profit is called an entrepreneur (Nbina, Viko and Birabil, 2010). According to professor Robert Hisrich (Hisrich, Peters and Shepherd, 2009, page 284) “*Entrepreneurship* is the process of creating something new that has value, and the entrepreneur- as a man who spends all his necessary time and effort, takes the full financial, psychological and social risk, receiving a reward money and satisfaction with the progress”. According to a study (Reynolds and White, 1997), each year between four and six percent of the working population in the United States take action to start a new venture, and about 40 percent of American adults experience spells of self employment in their lifetime. No matter how many entrepreneurs emerge, most do not succeed in creating lasting organizations (Aldrich and Martinez, 2001). Understanding how and why some entrepreneurs succeed remains a major challenge for the entrepreneurship research community.

For the case study presented, the students themselves act as nascent entrepreneur. Students utilize their technical skills and business knowledge in the product development and sale. In most cases, students themselves funded the projects. However, in some cases, the university provided partial funding to the less privileged students to meet the project expenses. The students were encouraged to develop and/or use their personal relationships with people having relevant technical expertise or successful entrepreneur who may provide useful business knowledge.

The aim of the research presented in this paper is to provide students the experience of a nascent entrepreneur. The concept of a “nascent entrepreneur” captures the flavor of the chaotic and disorderly founding process. Surveyed literature defines a nascent entrepreneur as someone who initiates serious activities that are intended to culminate in a viable business startup (Reynolds, 1994). The entrepreneurial cycle involves three transitions in entrepreneurial efforts: from an adult with a business idea to a nascent entrepreneur, to a fledgling firm, and finally, to an established new firm (Reynolds, 1994). The research conducted only covers the first transition i.e. the transition from an adult with a business idea to a nascent entrepreneur.

There are three essential elements for the success of nascent entrepreneurs (Aldrich and Martinez, 2001): human capital, financial capital and social capital. The creation of a new firm requires a certain amount of knowledge. A nascent entrepreneur’s formal education, previous experience, or informal training can help him in this context. Entrepreneurs also require financial capital in order to obtain the inputs (labour, raw materials, information, etc.) necessary for the production of their goods or services and to sustain them during the unavoidable period in which their efforts do not produce profits. Finally, entrepreneurs must also develop social networks to gain access to the information, knowledge, financial capital and other resources that they do not possess.

The risk factor and the desire to earn profit require that the entrepreneur develop a novel product. The product should satisfy the needs of potential customers and a reasonable product price must be set. Thus, the issues like innovation, customer satisfaction and pricing are some of the critical aspects of an entrepreneurial adventure. McGrath and McMillan (2000) have defined “entrepreneurial insight” as the crucial ingredient behind entrepreneurial success. *Entrepreneurial insight* is the ability to notice some favorable thing related to a business, which can provide superiority over the competitors.

Utilization of “Project Based Learning (PBL)” Concept and Comparison with Previous Work

In order to promote the concepts of business activity and entrepreneurship, instructor asked the students to develop a product by assigning projects to the students. Traditionally, many Computer Science &

Computer Engineering courses include projects as an essential component. PBL involves many phases like survey, design, development and testing but our innovation is to transform these projects into commercial product. Students interact with professionals to get ideas to win customer satisfaction. Many will not consider this entrepreneurial empowerment and economic activity as an earth shaking idea. However, it is a humble effort to find a solution to the contemporary issues causing unrest in several Middle Eastern and European countries through our higher educational institutions.

Historically, William Heard Kilpatrick (1918) first gave the idea of incorporation of projects in the curriculum. He used projects for simulating real life environment in a course. Since then project method or project based learning became widely applicable in a broad range of courses. Web contains huge repository of research documents involving project oriented learning. For instance, Yang and Wei (1999), Fernandez, and Williamson (2003) both used project-based approach to teach introductory programming courses. Fife (2010) has shown that projects are also useful for imparting authentic learning to students.

Project is an enthusiastic activity. Students try to supersede their peers by exhibiting creativity. This aspect of projects supplies nourishment to the creative thoughts and skills of students (Mirielli, Proorny and Buchan, 2010). In a previous iteration of the current work (author, year1; author, year2), creativity and innovation was also a dominant factor because it helps in winning the customer satisfaction. That is why (Pohl, 2011) regarded innovation as an important trait of entrepreneurs. Discussion about projects cannot end without highlighting the role of capstone courses. These courses have given new dimension to the concept of class projects by directing the projects towards benefiting the community. The novelty of this research and its previous iterations (Khan, 2010; Khan, 2011) is that they use projects to harvest business skills among the students. However, the current research also enlightens the entrepreneurial issues.

Currently, e-TShirts are equipped with a wide variety of components that range from simple LEDs to hi-fi sensors and micro-processors (Buechley, Elumeze, and Eisenberg, 2006). However, surveyed literature (Chenug, Chan, Ngai, & Lau, 2009; Crocket, Buchley, & Eisenberg, 2008; Schelhowe, Dittert, & Katterfeldt, 2009) shows that most of the research in e-Textiles spins around a micro-controller known as LilyPad Arduino. The peculiarity of this chip is that it is washable which makes it ideal for e-Textiles. Despite this great feature of LillyPad Arduino, (Ngai, G., Chan, F., Cheung, J. C. Y., Lau, W. W. Y., 2009) has questioned its integration in the circuit on the ground that developing expertise in programming this chip is time consuming. Another drawback associated with this chip is its unavailability in the local market. This requires its import from overseas at an exuberant cost. These bottlenecks resisted the use of LilyPad Arduino in the e-TShirt project and first author decided to develop e-TShirts with locally available components. The decision helped to keep the price of e-Shirt modest (in order to compete with the local brands). This is an important aspect of this research in the context of entrepreneurial insight.

The E-TShirt Projects

As discussed earlier, instructor assigned e-TShirt as a class project to the students. After receiving initial feedback from students, it became evident that the development of this product was not an easy task for the students because majority of them had never worked with electronic circuits. To tackle this situation, it was necessary to defer the e-TShirt project. Instead, students were asked to do some practice projects from the list given in link [<http://www.kpsec.freeuk.com/proj.htm>]. These projects in conjunction with weekly lab works were sufficient to give them the background knowledge required to undertake the e-TShirt project. e-TShirt project was finally assigned in week#9. Week#15 was set as the deadline for demonstration of e-TShirts in an exhibition event.

In the exhibition, students had to participate in a catwalk session by wearing the outfit. For this reason, students accentuated professional touch to their outfits so that it could fascinate the buyers. Thus, their success chiefly depended on a robust circuitry fastened to the e-TShirt and its innovative design. Along with this, they had to develop indigenous solution to deal with water proofing in the absence of LillyPad Arduino. One of the techniques to develop e-TShirt using LEDs as discussed by Post, Orth, Russo, Gershenfeld (2000) is E-broidery which did not function when wet but worked fine when dried again. Students tried to keep the circuit dry by using two techniques. The first technique formed an envelope using several layers of parachute cloth encasing the circuitry fixed on a thermo pole sheet. LEDs exposed through tiny holes from the envelope. Students also used water resistant glue to conceal any leaks produced by these holes. The other technique formed a detachable pocket for keeping the circuit, allowing the wearer to remove the pocket at the time of washing. Second option provided a versatile solution to the problem where as the first technique is good for hand-washing. Along with thermo pole sheets and basic electronic components, sewing stuff, LEDs, LED stripes, 555-timer IC, power source, and battery casing is required for the development of this project. Students had to submit a document illustrating the circuit diagram and the strategy to determine the price of their product by incorporating their expenses and the desired profit.

The E-TShirt Exhibition

Our nascent entrepreneurs needed a forum to enrich them with social capital. Indeed, through this forum they can interact with the local community to sell their e-TShirts. An exhibition event fulfilled this forum requirement. Benefits of exhibition event are multidimensional as discussed in author (year1). It can expedite students' efforts to complete their projects on time. It can also act as a hub for business activities.

Arrangement of exhibition event is a tough task. But students' zeal and interest undermined all hurdles as in the previous work (author, year1). However, in this work in the context of exhibition event, the organizing team decided to carry out some new experiments. In order to capitalize the benefits of exhibition, they promised of prizes that are more glamorous, along with media coverage and access to free video of the event. Students displayed the E-TShirts in a catwalk session in the university auditorium specially improvised for the event. Figure 1 depicts some e-TShirt designs presented during the exhibition.

Business activities were not limited to the students. Students' work also inspired the faculty to enter in small business ventures. On this basis business activities are grouped in to two types:

Student oriented Business Activities

These are the short-term activities generated by obtaining orders from the interested individuals and clients. The university acted as a guarantor on behalf of students to achieve customer's confidence.

Faculty oriented Business Activities

These are long-term business activities and may help in the establishment of Small and Medium Enterprises (SMEs).

More description about these two types of business activities is beyond the scope of this research.

Research Methodology

The key objective of the e-TShirt projects and exhibition was to indulge students in a useful business activity so that they may learn useful skills and attain valuable experience that is required to

start a new business. Another objective was to improve students' understanding and knowledge of the LDST course. After completion of the projects and ending of the exhibition, it is extremely important to check the success of this work by analyzing the achievements. In order to do this, we have decided to measure certain properties and features of the business activity and students' learning.

With the help of surveyed literature, we designed a questionnaire, containing numerous questions, for the purpose of data collection. The questions cover various business and students' learning issues. The following are the three question categories and a list of questions included for each of these categories:

1. The e-TShirt product
 - a. Do you easily get the components use in the product development from the local market?
 - b. Do you agree about the safety of the product? The e-TShirt contains electrical circuits with electrical current flowing in them. Does this pose any kind of risk to human body?
 - c. How much of your time and energy was consumed in the product development?
 - d. Are there ways and means to improve the e-TShirt product developed i.e. can you suggest any technical improvements?
 - e. Do you think that the product developed by you is innovative?
2. Students' learning / education
 - a. Do you think that learning entrepreneurial skills during term-time negatively affects students' performance in other courses due to substantial time consumption?
 - b. What impact does the experience of e-TShirt product development and its sale has on your performance in the course's exams, tests etc.?
 - c. Would you suggest permanent inclusion of entrepreneurial activities in computer engineering courses?
 - d. Do you think that imparting entrepreneurial education and entrepreneurial skills can help reduce unemployment problem?
3. Entrepreneurial /business issues
 - a. Customers of which age group and gender are expected to be most interested in buying the e-TShirt product?
 - b. What criteria did you use to set the price of e-TShirt product?
 - c. Are you satisfied with the quantity sold at the exhibition?
 - d. Did the experience of entrepreneurship gained, helped you improve your business skills and better understand the business process?
 - e. How many product units were developed?
 - f. Specify the count of units sold.
 - g. What was the price of a unit?
 - h. What was the cost of a unit

Most of the questions included are multiple-choice questions, while some (like question 1d) require textual answer. The e-TShirt product is considered to be feasible for business activity if the components needed are easily available, the product is safe, requires less time and energy to develop and there is some innovative feature(s) included in the product. Students also gave suggestions for technical improvements of this exercise.

We are confident that the product development and its sale should have a positive impact on students' performance in the course and should not affect students' performance in other courses. Students' opinion about whether they would recommend permanent inclusion of entrepreneurial activities in computer engineering courses along with their opinion about whether or not all this will ultimately result in unemployment reduction, are also taken through this questionnaire. If the students have enjoyed and

benefited from this entrepreneurial exercise, they would surely recommend inclusion of entrepreneurial activities in computer engineering courses. If the students think that the experience of entrepreneurship gained can substantially help them set up viable businesses, then they will most likely indicate that such an experience / training can (in the longer term) lead to reduction in unemployment.

We also collected some business data through the questionnaire. This includes data about the target population, pricing criteria and students' satisfaction regarding the quantity sold and the business skills learnt. The ultimate aim of any serious business activity is to earn profit. It is important that essential data required calculating the profit earned by students is gathered. For this reason, questions 3e – 3h have been included in the questionnaire. Answers to these questions will enable calculation of the profit earned by the students. If the students are successful in earning profit, then the business activity is a success.

A typical batch of computer engineering students at SSUET consists of around 300 – 400 students. 308 students took the LDST course in the semester when this exhibition and survey were conducted. Group size was three and each group had to produce a separate e-TShirt product. They were given ample time to design and develop their artifact. Several groups produced multiple units for each e-TShirt product designed. On the day of exhibition, the student groups presented and tried to sell their e-TShirt product in the exhibition held. Next day after the exhibition, each student group filled the questionnaire. This timing was chosen so that students are in a good position to answer the questions included in the questionnaire i.e. the timing was neither too early nor too late.

Results and Analysis

Hundred student groups filled the questionnaire and so the data set size is 100. Once the students returned the research questionnaire, the research results were compiled. Since the questionnaire questions can be divided in to three different categories, the results of the survey are also presented and analyzed in three different portions:

Results for questions in the “e-TShirt product” category

The aim of these questions is to get students' opinion on issues related to the product developed. Vast majority of student groups have indicated that the components used in the product development are easily available in the local market. **Error! Reference source not found.** illustrates this fact. The table's right column contains values for the number of student groups who support a particular point of view. Few student groups expressed difficulty in obtaining the required components while none indicated non-availability of components.

In the context of product safety issue, we were a bit surprised to learn that only 32 student groups consider their e-TShirt product completely safe, while 38 student groups have received feedback about some minor irritations caused due to use of thermo-pole sheets in their e-TShirt product. Other groups either have expressed the need for more safety testing or have not considered the safety issue at all. Thus, the safety issue needs more focus in future developments of this product.

The students have made a number of recommendations for improvements in the e-TShirt product. Some of the improvements suggested are:

1. More human comfort/protection required.
2. The garment should be waterproof / washable
3. Better quality circuits required.
4. More attractive e-TShirt designs required.

Some students have expressed the need for improvements but did not explicitly state what improvements they would like to see incorporated in to their e-TShirt product. Few students either did not answer this question or felt no need for any improvement. Around 80% of students' groups consider their product innovative. This is a good sign as this indicates students' confidence in their product. Time consumed in product development varies from a week to one and a half months. The great variation in time consumption is mainly because some students make a more concentrated effort and therefore require less time to complete while others make prolonged, less concentrated effort and therefore require more time to complete.

Results for questions in the "students' learning / education" category

The aim of questions in this category is to get students' opinion about the impact of this project on their learning/performance. We considered two factors (1) impact on students' performance in LDST course, and (2) impact on students' performance in other courses. Students' performance in other courses is also important and e-TShirt project should not undermine their status. The other two questions in this category require students to express their view on permanent inclusion of entrepreneurial activities in computer engineering courses on the pretext that it will lead to reduction in unemployment. The results obtained through these questions are encouraging. **Error! Reference source not found., Error! Reference source not found., Error! Reference source not found., and Error! Reference source not found.** show the precise results for each of these questions. Vast majority of student groups have indicated performance improvement in the LDST course's exam/test. Only small minority of student groups have reported no impact or negative impact of the entrepreneurial experience provided by the e-TShirt project. A reasonably high number of student groups stated that there is no negative impact on performance in other courses due to time consumption in the e-TShirt project. 93 out of 100 student groups have recommended permanent inclusion of entrepreneurial activities in computer engineering courses and all student groups think that imparting entrepreneurial skills and entrepreneurial education can help reduce unemployment problem (at least to some extent). All these are positive indicators and demonstrate students' high level of interest and appreciation of the e-TShirt project experience.

Results for questions in the "entrepreneurial / business issues" category

One question in this category requires student groups to state the class of customers (i.e. customers classified according to age group and gender) that were most interested in buying the e-TShirt product. A vast majority stated that male teenagers were most interested in buying e-TShirts, while some also indicated interest of female teenagers in the purchase of the e-TShirts.

Another question in this category was about the pricing criteria used. According to the responses obtained, the e-TShirt price depends upon the cost of the material and components used. Another critical factor was the human effort and the human creativity involved. Students kept the profit margin reasonably good but some price reduction was also possible if a customer was willing to purchase multiple units.

62% of the respondents have expressed complete satisfaction over the sale of their product at the exhibition, while 31% are satisfied to some extent and only 7% are completely unsatisfied. Similarly, 89% of the respondents believe that their business skills and understanding of the business process have substantially improved because of the entrepreneurial experience provided. Only 11% reported minor improvement while none reported no improvement. All these are encouraging results and provide rationale for inclusion of entrepreneurial activities in computer engineering / science courses. Table 4, shows that even though the number of units sold is substantially less than the number of units produced, the total profit earned is reasonably good. This was due to good profit margin on each unit

sold. If an entrepreneur with the necessary financial resources takes these results seriously, there is a possibility of launching a new company with a new brand.

Conclusion

Changes in the pedagogy towards teaching engineering students are long over due. Faculty can use the current economic crisis as a wakening call. They can assign innovative projects to their apprentice, exhorting them to unfold the underlying technology, currently in vogue. This exercise can also trigger business activity, causing students to learn entrepreneurial insight, as demonstrated in this research. Results indicate that students appreciated the efforts in this context. Other science related domains can also practice this technique by indulging students in marketable projects specific to their domains. Future work would aim at working out this technique to advance level courses of Bachelor of Science (BS) program.

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Appendix

Each Appendix appears on its own page.

Footnotes

¹Complete APA style formatting information may be found in the Publication Manual.

Table 1

Students' answers to the question: "Are the components used in the product development easily available in the local market?"

	No. of Student Groups
Not at all available	0
Difficulty available	14
Easily available	86

Table 2

Students' responses to the question: "What impact does the experience of e-TShirt product development and its sale has on your performance in the course's exams, tests etc.?"

	No. of Student Groups
The experience has improved exam/test performance	78
No impact	16
The experience has degraded exam/test performance	6

Table 3

Students' responses to the question: "Do you think that imparting entrepreneurial education and entrepreneurial skills can help reduce unemployment problem?"

	No. of Student Groups
Very much	33
To some extent	67
Not at all	0

Table 4

Summary of the entire business activity involving all the students' groups and customers

Total Number of Units Produced	671
Total Number of Units Sold	489
Total Production Cost of all the Units Produced	Rs. 234,850/-
Total Sale of all the Units Sold	Rs. 366,750/-

Figure Captions

Figure 1. Some e-TShirt designs presented during the exhibition

Figure 2. Students' responses to the question: "Do you think that learning entrepreneurial skills during term-time negatively affects students' performance in other courses?"

Figure 3. Students' responses to the question: "Would you suggest permanent inclusion of entrepreneurial activities in computer engineering courses?"

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